

Sparse Counterfactual Attribution of Crop-Yield Anomalies to Extreme-Weather Drivers

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Abstract

Machine-learning studies of crop yield usually optimize average predictive accuracy, but agricultural risk management also needs event-level explanations: why did a particular crop-state-year fall below its expected yield? We present Sparse Counterfactual Anomaly Attribution (SCAA), a model-based framework that detects detrended low-yield anomalies and explains them using sparse, physically grouped extreme-weather edits. Daily NASA POWER weather and USDA yield records are aggregated for four crops, twelve U.S. states, and 1990–2025. A forward-time ExtraTrees yield model attains $R^2 = 0.809$ for 2016–2021 and $R^2 = 0.808$ for 2019–2025. After crop-state detrending, 214 of 1,257 observations are flagged as low-yield anomalies. The main submission protocol is retrospective leave-one-event-year-out grouped SCAA: for every anomaly year, the residual model is fitted after excluding all rows from that year. Under this stricter protocol, the median recoverable fraction is 0.116 and 5.6% of anomalies (12 of 214) exceed a recovery threshold of $\phi \geq 0.5$. Residual-model validation is modest for all held-out event-year rows ($R^2 = 0.085$) and weak on anomaly rows ($R^2 = -5.152$), so SCAA is not framed as a high-skill residual forecasting system. Event-year consistency is reported as a limited sanity check rather than primary validation because broad heat/drought labels are also matched by simple null baselines. The contribution is a reproducible, conservative workflow for retrospective model-based weather-driver attribution on a detrended yield-residual scale, not a new yield prediction model.

Keywords: crop yield; extreme weather; counterfactual explanation; yield anomaly; interpretable machine learning; drought; heat stress.

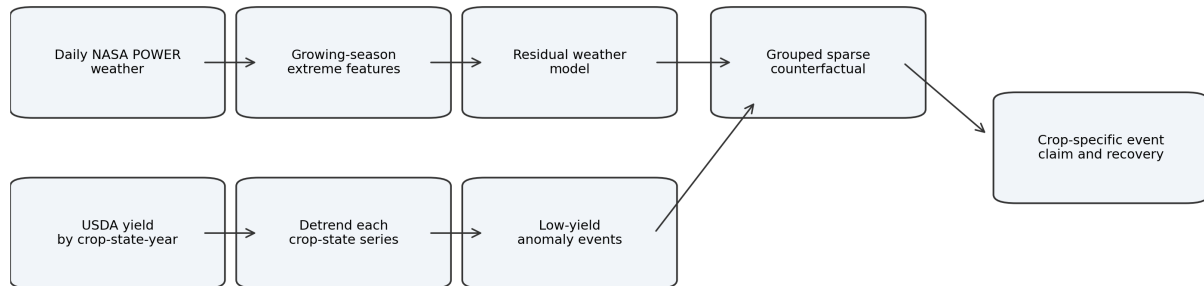
1 Introduction

Extreme weather can produce large agricultural losses, with drought, heat, excess water, and cold events affecting crop production at regional and global scales [7, 32, 34]. Standard crop-yield machine-learning workflows demonstrate that weather and climate variables can support accurate yield forecasting [5, 6, 11, 20], but prediction alone does not answer the event-level question needed for risk interpretation: which weather driver does the model identify as the strongest explanation for a specific low-yield crop-state-year?

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This paper reframes yield interpretation as anomaly attribution. We first separate long-run crop-state yield trend from short-run variation, following detrended yield residual analysis [8, 10, 22]. We then detect low-yield anomalies and ask a counterfactual question on the raw residual scale: what sparse, feasible change to a weather-driver group would move the predicted detrended residual toward a normal season? Unlike feature-importance methods such as SHAP or LIME [9, 23], this yields an event-level recovery quantity in t ha^{-1} and a crop-specific claim.

The contribution is not a new yield predictor. The contribution is a reproducible workflow that combines detrended yield anomalies, retrospective leave-one-event-year-out residual modelling, physically grouped extreme-weather drivers, and sparse counterfactual recovery. Figure 1 summarizes the pipeline.



SCAA explains detrended low-yield events with sparse, feasible, physically grouped weather changes.

Figure 1: SCAA workflow. Daily weather and yield records are converted into crop-state-year features, detrended anomalies, retrospective leave-one-event-year-out residual models, and grouped counterfactual recovery estimates.

2 Related Work

2.1 Crop-Yield Prediction and Weather Features

Large-scale crop-yield forecasting has been studied with random forests, neural networks, and other statistical or machine-learning approaches [5, 20, 25]. In-season forecasting under small-data conditions is also active, especially for grain systems [11]. These studies support the predictive modeling layer of our work, but they mainly optimize forecasting skill rather than event-level explanation.

2.2 Extreme-Weather Yield Anomalies

Prior work links yield anomalies to climate extremes, including heatwaves, drought, water excess, hot-dry compound events, and crop-specific weather relationships [4, 27, 32, 34]. Stage-specific weather effects are also important because temperature, rainfall, frost, and radiation can matter differently during crop development [24]. These studies motivate our driver groups: heat, drought/dry-spell, frost/cold, excess rainfall, and radiation.

2.3 Counterfactual and Event-Level Attribution

Counterfactual explanations ask for minimal changes that would alter a model outcome [33]. Diverse and feasible counterfactual methods emphasize proximity, sparsity, plausibility, and actionable

constraints [12, 21, 29, 31]. Climate-event attribution literature also emphasizes careful event definition and cautious interpretation [3, 19, 30]. We use the counterfactual form for model-based explanation of yield residuals and report consistency with known stress groups rather than formal event-attribution effects.

3 Data

The empirical unit is a crop-state-year. The processed frame contains 1,257 observations for Barley, Canola, Oats, and Wheat across twelve U.S. states from 1990–2025. Yields are in t ha^{-1} and are derived from USDA NASS yield records [28]. Weather features are derived from NASA POWER daily records [13] and include rainfall totals, dry spells, heavy-rain events, heat days, heatwaves, growing-degree days, frost/cold metrics, and radiation summaries.

The weather extraction uses one representative latitude-longitude coordinate per state. Thus, all crops in a state share the same daily weather series before crop-specific growing-season windows are applied. The design is transparent and reproducible, but it should be interpreted as representative-coordinate state weather rather than a crop-area-weighted state average or a within-state grid average. Growing season windows are crop- and state-aware, with spring and winter windows represented in the processed frame.

Table 1: Dataset summary by crop.

crop	observations	states	years	windows	anomalies
Barley	283	10	1990-2025	spring	52
Canola	133	6	1991-2025	spring	21
Oats	416	12	1990-2025	spring	71
Wheat	425	12	1990-2025	spring, winter	70

Table 2: Extreme-weather driver groups used by grouped SCAA.

driver_group	description
heat	High-temperature and heatwave exposure
drought	Rainfall deficit, dry days, and dry-spell persistence
frost_cold	Cold and frost exposure
excess_rain	Wetness and heavy-rainfall intensity
radiation	Seasonal solar-radiation anomaly

Figure 2 shows crop coverage and crop-state density.

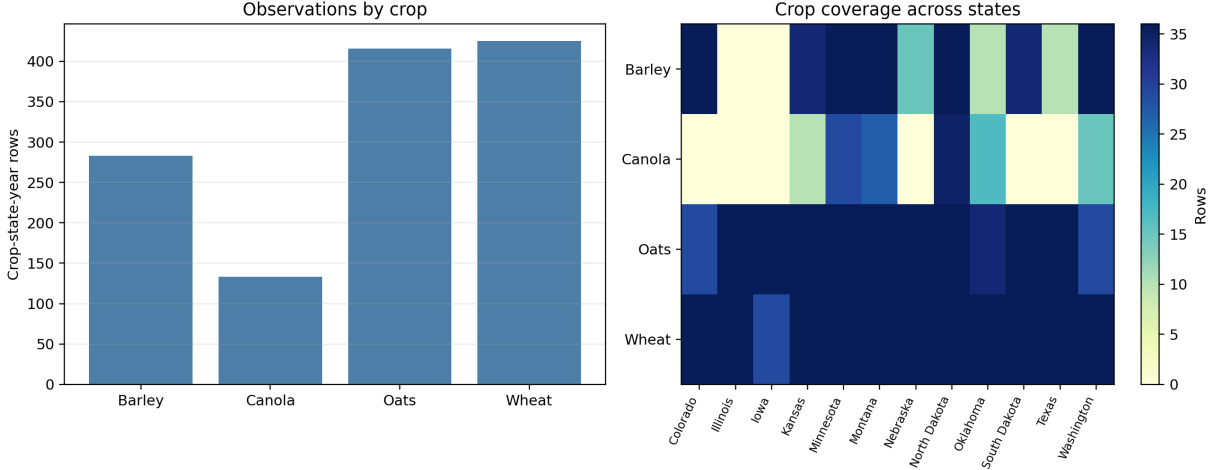


Figure 2: Data coverage by crop and state.

4 Method

4.1 Yield Skill Baseline

We retain a forward-time yield model to verify that the weather and crop-state features contain predictive signal. Numeric variables are median-imputed and standardized; categorical crop and state identifiers are one-hot encoded. An ExtraTrees regressor is trained under two forward-time splits: train through 2015 and test 2016–2021; then train through 2018 and test 2019–2025.

4.2 Detrending and Anomaly Screening

For each crop-state series, a linear trend is fit to yield:

$$y_{c,s,t} = \alpha_{c,s} + \beta_{c,s}t + \epsilon_{c,s,t}. \quad (1)$$

The raw detrended residual is

$$r_i = y_i - y_i^{trend}, \quad (2)$$

with unit t ha^{-1} . The standardized residual z_i is used only to screen low-yield anomalies. An observation is flagged when $z_i < -1.0$. We use the term “low-yield anomaly” rather than “extreme anomaly” because this threshold is meant to capture a broad set of below-trend events for attribution.

4.3 Retrospective Leave-One-Event-Year-Out Grouped SCAA

The main submission method is retrospective leave-one-event-year-out grouped SCAA. For each anomaly year t^* , we train a residual model f_{-t^*} on all crop-state-year rows except those from t^* . This prevents the model from directly learning the same event year being explained. It is not a fully prospective forecast, however, because observations after t^* may remain in the training set. The target is the raw detrended yield residual `trend_residual_t_ha`, not the standardized residual. The fitted model maps weather features, crop, and state to predicted residual $\hat{r}_i$ on the same t ha^{-1} scale.

For an anomaly i , grouped SCAA evaluates driver groups one at a time. For a group g , all weather features in that group are moved toward their observed median for the same region-window

support, giving a diagnostic edited feature vector $\mathbf{x}_i^{cf(g)}$. The observed detrended yield shortfall is

$$\Delta_i = \max(0, y_i^{trend} - y_i). \quad (3)$$

The recovered yield gap for the best group edit is

$$\text{recovered_gap}_i = \min \left(\max(0, \hat{r}_i^{cf} - \hat{r}_i), \Delta_i \right), \quad (4)$$

and the recoverable fraction is

$$\phi_i = \begin{cases} \text{recovered_gap}_i / \Delta_i, & \Delta_i > 0, \\ 0, & \Delta_i = 0. \end{cases} \quad (5)$$

A crop-state-year is labelled weather-driven when $\phi_i \geq 0.5$.

The grouped edit is a diagnostic sparse edit designed to isolate one physical driver group. It is not assumed to be a complete season that occurred historically. For that reason, we also report observed-analog counterfactuals: real non-anomalous seasons from the same crop-state, or the same region-window fallback, used as a plausibility check for weather-vector replacement.

4.4 Residual-Model Validation

Because grouped SCAA operates on the raw detrended residual scale, we separately evaluate the residual model used for attribution. We report R^2 , RMSE, and MAE for `trend_residual_t_ha` under a forward-time residual split and under the same retrospective leave-one-event-year-out protocol used by grouped SCAA. These metrics do not replace the forward-time yield model; they verify how much residual-scale signal is available before diagnostic counterfactual recovery is interpreted.

4.5 Robustness and Extensions

We include four robustness or extension layers. First, in-sample grouped SCAA is kept as an exploratory comparison to show the effect of removing year holdout. Second, observed-analog counterfactuals test whether recovery is possible using historically observed normal seasons. Third, anomaly-threshold and detrending sensitivity tables check whether the anomaly set is an artifact of a single screening choice. Fourth, crop-specific vulnerability profiles stress-test each crop only on observed crop-state-year combinations. An early-warning extension trains anomaly classifiers from early and early-plus-mid season weather features, with calibration-derived uncertainty bands [1, 2, 26].

5 Results

5.1 Yield Skill and Anomalies

The yield model reproduces strong forward-time predictive skill (Table 3; Figure 3). The 2016–2021 split attains $R^2 = 0.809$ and RMSE 0.525 t ha^{-1} ; the 2019–2025 split attains $R^2 = 0.808$ and RMSE 0.536 t ha^{-1} . After detrending, 214 of 1,257 observations are flagged as low-yield anomalies. The anomaly timeline highlights 2012, 2021, and 2022 as event years used for the consistency check (Figure 4).

Table 3: Forward-time yield performance and anomaly count.

evaluation	test_years	n_test	r2	rmse_t_ha
test_2016_2021	2016-2021	209	0.809	0.525
test_2019_2025	2019-2025	236	0.808	0.536
window_spring_2016_2021	2016-2021	164	0.842	0.475
window_winter_2016_2021	2016-2021	45	0.543	0.677
low_yield_anomalies	1990-2025	1257		214 anomalies

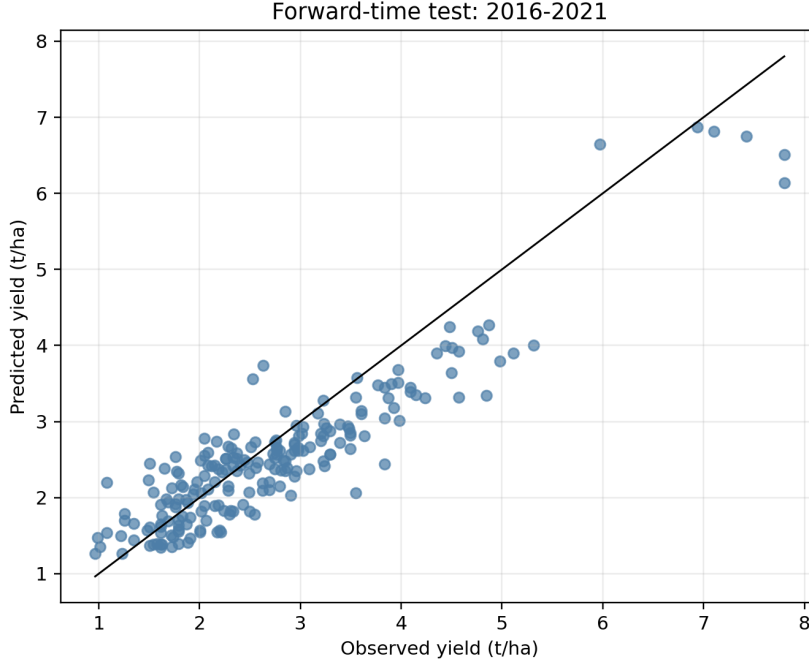


Figure 3: Observed versus predicted yield under the forward-time 2016–2021 test.

5.2 Residual-Model Validation

Table 4 reports the predictive skill of the residual model used by grouped SCAA. The forward-time residual split has $R^2 = 0.093$, RMSE 0.431 t ha^{-1} , and MAE 0.326 t ha^{-1} . Under retrospective leave-one-event-year-out evaluation across all held-out rows, $R^2 = 0.085$, RMSE is 0.369 t ha^{-1} , and MAE is 0.284 t ha^{-1} . On the anomaly rows alone, R^2 is negative ($R^2 = -5.152$), indicating that anomaly-level residual magnitudes are difficult to predict directly. We therefore interpret SCAA as a diagnostic recovery framework with bounded counterfactual edits, not as a high-skill residual forecasting model.

5.3 Method Comparison

Table 5 and Figure 5 compare methods by raw metrics only. The retrospective leave-one-event-year-out method is more conservative than the in-sample grouped method: median recoverable fraction decreases from 0.589 to 0.116, and the weather-driven rate decreases from 73.4% to 5.6% (12 of

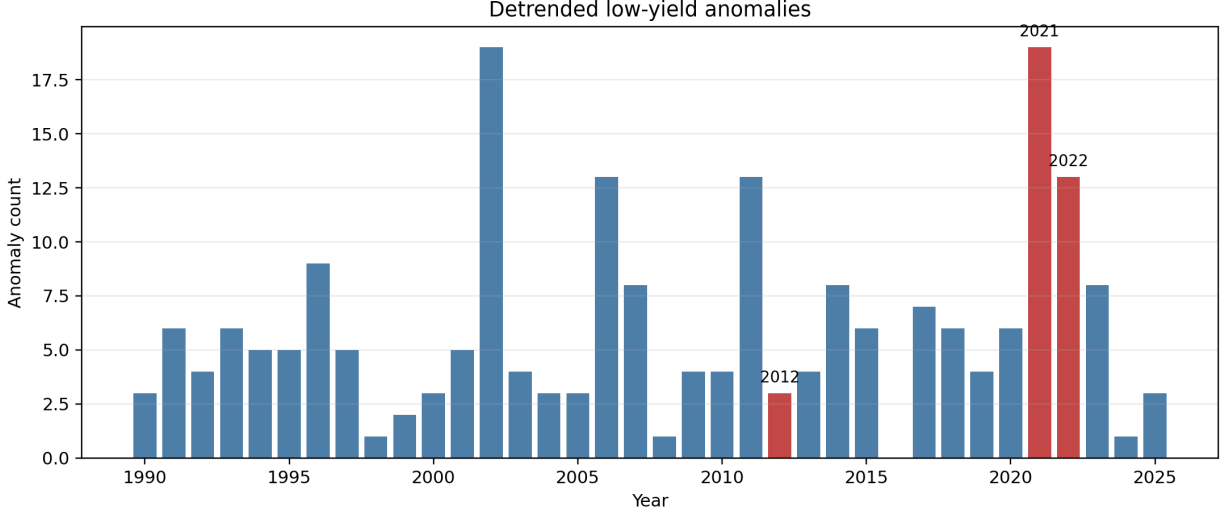


Figure 4: Low-yield anomaly counts by year, with 2012, 2021, and 2022 highlighted.

Table 4: Residual-model validation for the attribution target.

protocol	held_out_unit	n_eval	r2	rmse_t_ha	mae_t_ha
Forward-time residual	2016-2025	343	0.093	0.431	0.326
Leave-one-event-year-out residual	All held-out rows	1222	0.085	0.369	0.284
Leave-one-event-year-out residual	Anomaly rows	214	-5.152	0.562	0.496

214 anomalies). This reduction is expected because each event year is excluded before attribution. We therefore use leave-one-event-year-out grouped SCAA for the main submission claims, while the in-sample grouped result is treated as exploratory. Observed-analog counterfactuals show larger recovery because they replace the full weather vector with a real normal season; they are used as a plausibility comparison rather than the main sparse explanation.

Table 5: Attribution-method comparison by raw metrics; composite scores are not used in the manuscript.

method	median_recoverable_fraction	weather_driven_rate	event_expected_match_rate
Raw-yield SCAA	0.379	0.262	0.829
Residual-target SCAA	0.482	0.472	0.629
Grouped-driver SCAA	0.589	0.734	0.886
Leave-one-event-year-out grouped SCAA	0.116	0.056	0.914
Observed-analog counterfactual	0.853	0.972	0.971

5.4 Leave-One-Event-Year-Out Grouped Attribution

Retrospective leave-one-event-year-out grouped SCAA assigns a recovery fraction and dominant driver group to each low-yield anomaly (Figure 6). The median recoverable fraction is 0.116 and the mean is 0.157. Drought/dry-spell stress is the most frequent driver group, followed by heat exposure, excess rainfall or wetness, frost/cold stress, and radiation. The top crop-specific event claims are shown in Table 6. For example, North Dakota Canola in 2012 is attributed to drought/dry-spell stress, with a modelled recovery of 0.183 t ha^{-1} , or 64.0% of the observed detrended yield shortfall.

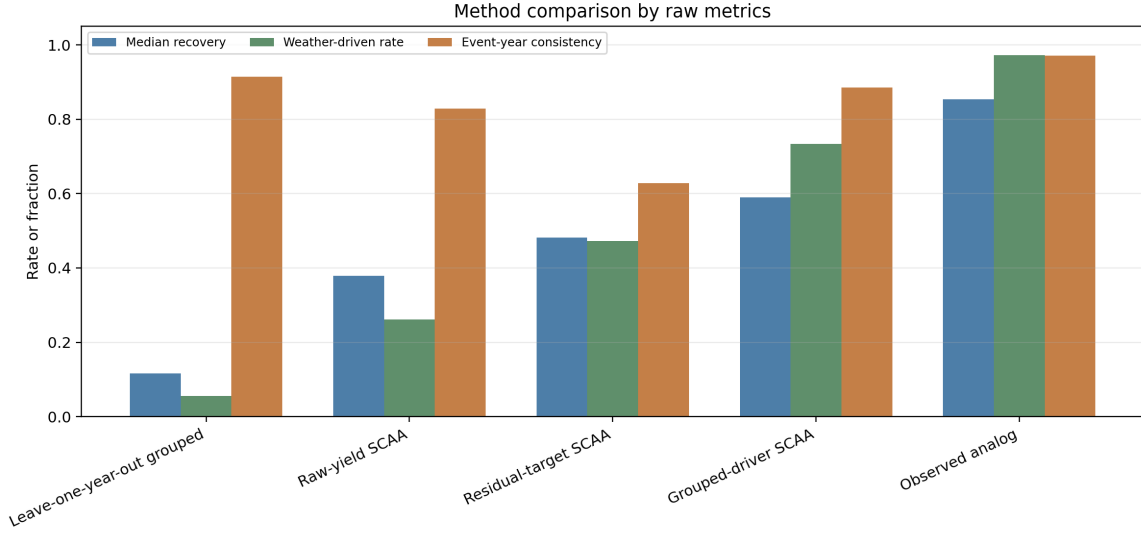


Figure 5: Method comparison by raw metrics with readable method labels. No composite score is used in the main manuscript.

Table 6: Top retrospective leave-one-year-out grouped-SCAA crop-region-year attribution claims.

crop	region	year	driver_group	dominant_feature	yield_gap_t_ha	recovered_gap_t_ha	recoverable_fraction
Oats	South Dakota	2006	drought	rain_mean	0.381	0.273	0.717
Barley	Washington	2015	drought	hot_dry_days_30_1mm	0.862	0.598	0.693
Wheat	Washington	1992	drought	max_dry_spell_1mm	0.579	0.39	0.672
Wheat	Texas	2009	drought	dry_spell_events_14d	0.41	0.273	0.666
Canola	North Dakota	2012	drought	hot_dry_days_30_1mm	0.285	0.183	0.64

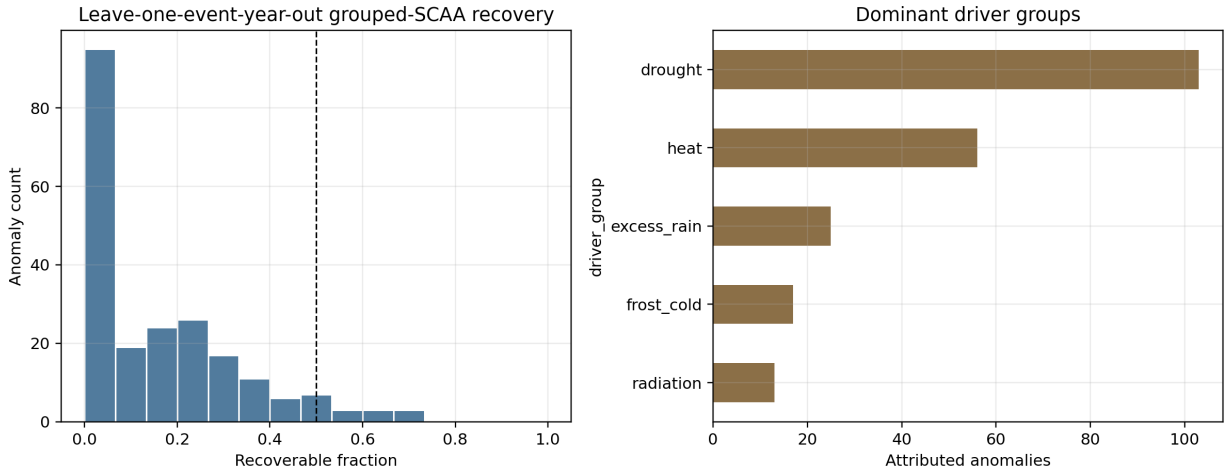


Figure 6: Retrospective leave-one-event-year-out grouped-SCAA recoverable fraction and dominant driver-group frequency.

5.5 Event-Year Consistency Check

We pre-specified broad expected stress groups for 2012, 2021, and 2022 using external NOAA/NCEI, National Weather Service, and Drought.gov/NIDIS evidence [14–18]. This check asks whether the SCAA dominant groups are consistent with those external stress categories; it does not validate formal event drivers. Retrospective leave-one-event-year-out grouped SCAA matches the pre-specified expected groups in 91.4% of event-year rows: 100% in 2012, 100% in 2021, and 76.9% in 2022.

The null baselines in Table 7 show why this result should be treated cautiously.

Table 7: Null baselines for event-year consistency checks.

Method	Expected match rate	Median recovery	n event rows
Always drought	1.000	–	35
Always heat	1.000	–	35
Most frequent driver (heat)	1.000	–	35
Driver-frequency random	0.896	–	35
Leave-one-event-year-out grouped SCAA	0.914	0.182	35

The first three baselines show that the broad heat/drought labels are easy to match. Therefore, event-year consistency is retained only as a sanity check that SCAA does not contradict known stress categories, not as primary validation or evidence of superiority over trivial labels.

The detailed event-evidence table and event-year consistency panel are provided in the Supplementary Material to keep the main manuscript focused on the core diagnostic result.

5.6 Crop-Specific Vulnerability and Early Warning

The vulnerability profile answers which weather driver most reduces which crop in the fitted residual model, evaluated only on observed crop-state-year combinations. Barley is most sensitive to heat exposure, with median residual effect -0.134 t ha^{-1} under adverse observed extremes; Wheat is most sensitive to drought/dry-spell stress, with median effect -0.103 t ha^{-1} ; and Canola is most sensitive to heat exposure, with median effect -0.086 t ha^{-1} .

Table 8: Crop-specific vulnerability profiles under adverse observed weather extremes.

crop	driver_group	median_effect_t_ha	effect_direction	states_most_sensitive
Barley	heat	-0.134	adverse	Oklahoma (-0.253); South Dakota (-0.222); Kansas (-0.191)
Wheat	drought	-0.103	adverse	Washington (-0.378); Colorado (-0.327); Oklahoma (-0.282)
Canola	heat	-0.086	adverse	Kansas (-0.245); Minnesota (-0.167); Oklahoma (-0.142)
Barley	drought	-0.057	adverse	Washington (-0.249); North Dakota (-0.165); Nebraska (-0.143)
Canola	drought	-0.049	adverse	Washington (-0.175); North Dakota (-0.125); Montana (-0.116)
Wheat	heat	-0.041	adverse	South Dakota (-0.224); Montana (-0.173); Minnesota (-0.110)
Oats	heat	-0.034	adverse	South Dakota (-0.162); Colorado (-0.128); Nebraska (-0.118)

Early-warning metrics and the early-versus-mid-season comparison figure are provided in the Supplementary Material. The extension is exploratory because anomaly labels and detrending are defined retrospectively in this version.

6 Discussion

The revised results support a conservative paper framing. The retrospective leave-one-event-year-out SCAA protocol produces lower recovery than the in-sample grouped method, but it better

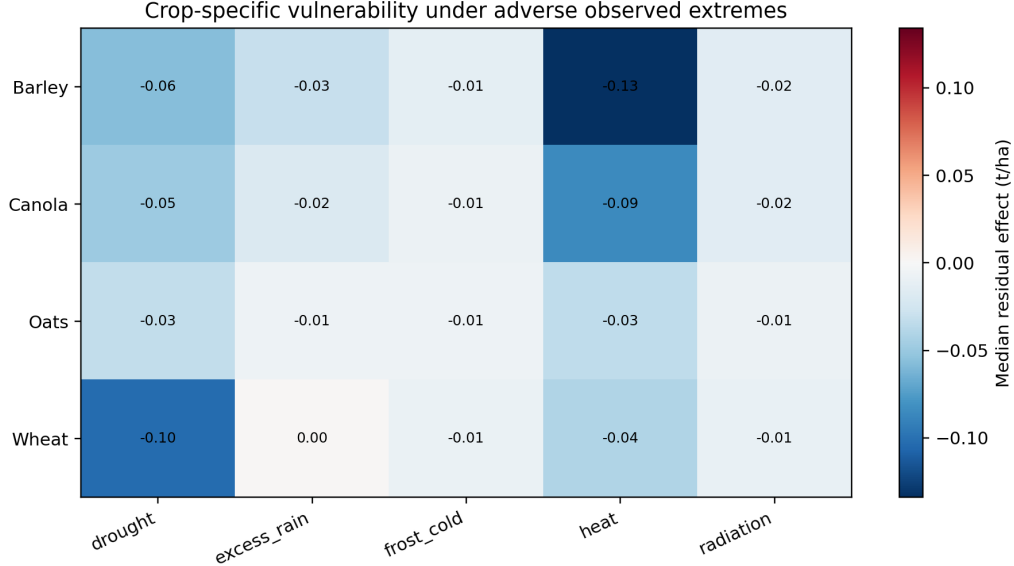


Figure 7: Crop-specific vulnerability profiles under adverse observed driver-group extremes.

addresses external-by-year attribution concerns because the event year is absent from the residual-model fit. Its strongest contribution is therefore not high recovery for every anomaly; it is a transparent event-level protocol that reports crop, state, year, driver group, dominant feature, recovered t ha^{-1} , and the residual-scale recovery formula.

The residual model itself is weak on anomaly rows ($R^2 = -5.152$), so high-recovery cases should be read as conservative diagnostic explanations within the fitted model rather than as evidence of broad residual-forecasting skill.

The threshold sensitivity analysis shows that the anomaly count changes from 214 at $z < -1.0$ to 85 at $z < -1.5$ and 30 at $z < -2.0$, while the leave-one-event-year-out median recoverable fraction remains close to 0.12 across thresholds. Detrending robustness is moderate: quadratic detrending has Jaccard overlap 0.774 with the linear anomaly set, while a centered seven-year rolling baseline has overlap 0.550. We keep linear detrending because it is transparent and low-parameter, but report these alternatives to make the screening choice explicit.

There are several limitations. The counterfactuals are model-internal predictive associations and should not be read as formal climate-event attribution. State-level representative-coordinate weather hides within-state variation in weather, irrigation, soil, cultivar, and management. The detrending step is also estimated over the full 1990–2025 retrospective crop-state series; an operational early-warning deployment would require rolling or past-only trend estimation before anomaly screening. The grouped edit changes several related features toward medians and may not represent a physically complete weather season; observed-analog counterfactuals partly address this by using historical seasons that actually occurred. Finally, the early-warning extension uses realized early-season weather, not operational seasonal forecasts.

7 Conclusion

SCAA detects detrended low-yield anomalies and explains them through sparse, physically grouped weather counterfactuals on a raw residual scale. On U.S. Barley, Canola, Oats, and Wheat across 1990–2025, the retrospective leave-one-event-year-out grouped-SCAA protocol produces conservative crop-state-year claims such as drought/dry-spell stress in North Dakota Canola 2012, with

explicit recovered $t \text{ ha}^{-1}$ and recoverable fractions. The result is a reproducible bridge between crop-yield prediction, interpretable machine learning, and cautious diagnostic event-level weather-driver analysis.

Additional driver-feature, robustness, event-consistency, and early-warning materials are provided in the Supplementary Material.

References

- [1] Weston Anderson, Shraddhanand Shukla, James Verdin, et al. Preseason maize and wheat yield forecasts for early warning of crop failure. *Nature Communications*, 15:7262, 2024. doi: 10.1038/s41467-024-51555-8.
- [2] Mohamed Farag, Ahmed Emam, Johannes Leonhardt, and Ribana Roscher. Enhancing decision support in crop production: Analyzing conformal prediction for uncertainty quantification. *Computers and Electronics in Agriculture*, 237:110559, 2025. doi: 10.1016/j.compag.2025.110559.
- [3] Alexis Hannart, Judea Pearl, Friederike E. L. Otto, Philippe Naveau, and Michael Ghil. Causal counterfactual theory for the attribution of weather and climate-related events. *Bulletin of the American Meteorological Society*, 97:99–110, 2016. doi: 10.1175/BAMS-D-14-00034.1.
- [4] Matti Heino et al. Increased probability of hot and dry weather extremes during the growing season threatens global crop yields. *Scientific Reports*, 2023. doi: 10.1038/s41598-023-29378-2.
- [5] Saeed Khaki and Lizhi Wang. Crop yield prediction using deep neural networks. *Frontiers in Plant Science*, 2019. doi: 10.3389/fpls.2019.00621.
- [6] Guoyong Leng and Jim W. Hall. Predicting spatial and temporal variability in crop yields: an inter-comparison of machine learning, regression and process-based models. *Environmental Research Letters*, 15:044027, 2020. doi: 10.1088/1748-9326/ab7b24.
- [7] Corey Lesk, Pedram Rowhani, and Navin Ramankutty. Influence of extreme weather disasters on global crop production. *Nature*, 529:84–87, 2016. doi: 10.1038/nature16467.
- [8] Jin Lu, Gregory J. Carbone, and Peng Gao. Detrending crop yield data for spatial visualization of drought impacts in the united states, 1895–2014. *Agricultural and Forest Meteorology*, 237–238:196–208, 2017. doi: 10.1016/j.agrformet.2017.02.001.
- [9] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, 2017.
- [10] H. Meng et al. Performances of different yield-detrending methods in assessing drought and flooding impacts on crop yield. *Agricultural Water Management*, 296:108795, 2024. doi: 10.1016/j.agwat.2024.108795.
- [11] Michele Meroni, Francois Waldner, Lucia Seguíni, Herve Kerdiles, and Felix Rembold. Yield forecasting with machine learning and small data: What gains for grains? *Agricultural and Forest Meteorology*, 308–309:108555, 2021. doi: 10.1016/j.agrformet.2021.108555.
- [12] Ramaravind K. Mothilal, Amit Sharma, and Chenhao Tan. Explaining machine learning classifiers through diverse counterfactual explanations. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 2020. doi: 10.1145/3351095.3372850.

- [13] NASA Prediction Of Worldwide Energy Resources Project. Power data access viewer and api, 2025. URL <https://power.larc.nasa.gov/data-access-viewer/>. Accessed 2026-06-15.
- [14] NOAA National Centers for Environmental Information. Monthly drought report: Annual 2012, 2013. URL <https://www.ncei.noaa.gov/access/monitoring/monthly-report/drought/201213>. Accessed 2026-06-15.
- [15] NOAA National Centers for Environmental Information. Monthly drought report: August 2022, 2022. URL <https://www.ncei.noaa.gov/access/monitoring/monthly-report/drought/202208>. Accessed 2026-06-15.
- [16] NOAA National Centers for Environmental Information. Monthly drought report: Annual 2022, 2023. URL <https://www.ncei.noaa.gov/access/monitoring/monthly-report/drought/202213>. Accessed 2026-06-15.
- [17] NOAA National Integrated Drought Information System. 2020-2021 drought in the u.s. northern plains and canadian prairies, 2022. URL <https://www.drought.gov/documents/2020-2021-drought-us-northern-plains-and-canadian-prairies>. Accessed 2026-06-15.
- [18] NOAA National Weather Service. 2012 drought, 2012. URL https://www.weather.gov/iwx/2012_drought. Accessed 2026-06-15.
- [19] Friederike E. L. Otto. Attribution of weather and climate events. *Annual Review of Environment and Resources*, 42:627–646, 2017. doi: 10.1146/annurev-environ-102016-060847.
- [20] D. Paudel et al. Machine learning for large-scale crop yield forecasting. *Agricultural Systems*, 187:103016, 2021. doi: 10.1016/j.agsy.2020.103016.
- [21] Rafael Poyiadzi, Kacper Sokol, Raul Santos-Rodriguez, Tijl De Bie, and Peter Flach. Face: Feasible and actionable counterfactual explanations. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 2020. doi: 10.1145/3375627.3375850.
- [22] Deepak K. Ray, James S. Gerber, Graham K. MacDonald, and Paul C. West. Climate variation explains a third of global crop yield variability. *Nature Communications*, 6:5989, 2015. doi: 10.1038/ncomms6989.
- [23] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. Why should i trust you? explaining the predictions of any classifier. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016. doi: 10.1145/2939672.2939778.
- [24] Florian Schierhorn, Michael Hofmann, Taras Gagalyuk, Iryna Ostapchuk, and Daniel Muller. Machine learning reveals complex effects of climatic means and weather extremes on wheat yields during different plant developmental stages. *Climatic Change*, 169:39, 2021. doi: 10.1007/s10584-021-03272-0.
- [25] Mohsen Shahhosseini, Guiping Hu, Isabella Huber, and Sotirios V. Archontoulis. Coupling machine learning and crop modeling improves crop yield prediction in the us corn belt. *Scientific Reports*, 2021. doi: 10.1038/s41598-020-80820-1.
- [26] G. Singh et al. Uncertainty quantification for probabilistic machine learning in earth observation using conformal prediction. *Scientific Reports*, 2024. doi: 10.1038/s41598-024-65954-w.

- [27] H. Sjulgard, T. Keller, G. Garland, and T. Colombi. Relationships between weather and yield anomalies vary with crop type and latitude in sweden. *Agricultural Systems*, 211:103757, 2023. doi: 10.1016/j.agsy.2023.103757.
- [28] United States Department of Agriculture, National Agricultural Statistics Service. Quick stats database and api, 2025. URL https://www.nass.usda.gov/Quick_Stats/. Accessed 2026-06-15.
- [29] Berk Ustun, Alexander Spangher, and Yang Liu. Actionable recourse in linear classification. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 2019. doi: 10.1145/3287560.3287566.
- [30] Geert Jan van Oldenborgh et al. Pathways and pitfalls in extreme event attribution. *Climatic Change*, 166:13, 2021. doi: 10.1007/s10584-021-03071-7.
- [31] Sahil Verma et al. Counterfactual explanations and algorithmic recourses for machine learning: A review. *ACM Computing Surveys*, 2024. doi: 10.1145/3677119.
- [32] Elisabeth Vogel et al. The effects of climate extremes on global agricultural yields. *Environmental Research Letters*, 2019. doi: 10.1088/1748-9326/ab154b.
- [33] Sandra Wachter, Brent Mittelstadt, and Chris Russell. Counterfactual explanations without opening the black box. *Harvard Journal of Law and Technology*, 31:841–887, 2018. doi: 10.2139/ssrn.3063289.
- [34] Matteo Zampieri, Andrej Ceglar, Frank Dentener, and Andrea Toretì. Wheat yield loss attributable to heat waves, drought and water excess. *Environmental Research Letters*, 2017. doi: 10.1088/1748-9326/aa723b.